

Hunter Browning

Phoenix, AZ

R&D Hardware Engineer | Mixed-Signal PCB Design | Interfaces | Verification

Hardware design engineer with experience developing embedded mixed-signal PCB hardware, optoelectronic/metrology systems, prototype debug, EMI/ESD mitigation, and verification test planning. Additional high-reliability aerospace experience supporting ECN/redline workflows, substantiation documentation, requirements closure, and configuration-controlled engineering processes.

Professional Experience

Honeywell – Aerospace // Electrical Engineer

Oct. 2025 – Present

Completed engineering change deliverables for obsolescence-driven updates to high-reliability hardware across legacy commercial and military programs.

- Acted as a product engineer for component/assembly upgrades, owned the process through formal ECN/redline workflows, and maintained traceability/configuration integrity in Siemens TeamCenter.
- Authored and reviewed substantiations (analysis reports, test procedures, white papers) to support qualification impacts and technical reviews.
- Coordinated requirements and verification closure using JIRA and IBM DOORS, improving review readiness and reducing rework for update cycles.

KLA – MicroSense R&D // Hardware Design Engineer

Feb. 2024 – Oct. 2025

Designed embedded mixed-signal PCBs for optoelectronic wafer metrology and nanotopography systems.

- Released 12+ CCAs; improved subsystem performance through analog/digital partitioning, noise-aware layout, and thermal optimization.
- Improved robustness in high-EMI assemblies by co-designing enclosure/PCB interfaces (shielding, grounding, ESD), reducing susceptibility and downtime.
- Authored verification test plans and debugged 50+ prototype issues across analog, digital, and interconnect domains.

Education

Northern Arizona University – BSc. Computer Hardware Engineering

Dec. 2023

- IEEE Chapter President – Led technical workshops, hackathons, industry events, and field trips
- Lab Equipment Manager – Maintained/repaired test equipment, 3D printers, and lab inventory
- Teaching Assistant – Mentored 12+ scrum teams on multi-semester engineering projects

Georgia Military College – ASc. Mathematics

Dec. 2017

- Peer Tutor – Math, physics, chemistry

Skills

- Design – Mixed-signal, low-noise analog, power, grounding/shielding, EMI/ESD, bring-up/debug
- Test – Oscilloscope, logic analyzer, DMM, function generator, solder/rework, failure isolation
- EDA and CAD – KiCad, SolidWorks, Allegro X, PSPICE, LTSpice, DFM/DFX
- Programming – C, C++, C#, MATLAB, Python, Visual Basic
- Interfaces – SPI, I2C, JTAG, USB, DDR4, RJ45, RS-485/422, ARINC 664